

# Sparse Recovery and Compressed Sensing

## A Model-Based Deep Learning Approach

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*Core question: How far can model-based structure take us — and when do we need data-driven flexibility?*

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*Based on: Chen, Liu, Wang, Yin — "Hyperparameter Tuning is All You Need for LISTA" • NeurIPS 2021*

# Seminar Roadmap

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## Background

Motivation, LASSO,  
ISTA / FISTA

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## Unrolling

LISTA, ALISTA,  
HyperLISTA

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## Our Study

Scalar variants +  
training methods

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## Results

Sparse recovery &  
design space

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## Images

MNIST & FashionMNIST  
CS stress tests

Goal: systematically compare how much learning is needed when moving from ideal sparse signals to real compressed-sensing images.

### Synthetic signals

$m=250$ ,  $n=500$ ,  $s=50$   
Gaussian  $A$ , noiseless

### Trained models

LISTA, ALISTA, HyperLISTA  
+ 3 scalar designs (ours)

### Image CS

MNIST & FashionMNIST pixel domain  
ratios  $\in \{0.25, 0.5\}$

# Motivation & Problem Formulation

Sparse recovery appears everywhere:

| Application        | Measurements          | Goal            |
|--------------------|-----------------------|-----------------|
| MRI acceleration   | k-space ( $m \ll n$ ) | Image pixels    |
| Radar              | Compressed snapshots  | Reflectivity    |
| Hyperspectral      | Spectral bands        | Spatial cube    |
| Compressed sensing | Random projections    | Original signal |

Why tractable? Natural signals have only  $s \ll n$  non-zero components.

**LASSO — the optimization backbone:**

$$\hat{x} = \arg \min_x \frac{1}{2} \|b - Ax\|_2^2 + \lambda \|x\|_1$$

LASSO = Least Absolute Shrinkage and Selection Operator (Tibshirani, 1996)

Measurement model

$$b = Ax^* + \varepsilon$$

$A \in \mathbb{R}^{m \times n}$ ,  $m \ll n$

$x^*$ :  $s$ -sparse signal

$\varepsilon$ : additive Gaussian noise

Metrics: NMSE (dB) for synthetic signals;  
NMSE + PSNR + SSIM for image CS.

# Classical Baselines: ISTA and FISTA

ISTA — proximal gradient descent on the LASSO objective:

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta} \left( \mathbf{x}^{(k)} + (1/L) A^T (b - A \mathbf{x}^{(k)}) \right), \quad L = \|A\|_2^2, \quad \theta = \lambda / L$$

FISTA — ISTA with Nesterov momentum,  $O(1/k^2)$  convergence vs  $O(1/k)$ :

$$\mathbf{y}^{(k)} = \mathbf{x}^{(k)} + ((t_k - 1) / t_{k+1}) (\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}), \quad \mathbf{x}^{(k+1)} = \mathcal{S}_{\theta} \left( \mathbf{y}^{(k)} + (1/L) A^T (b - A \mathbf{y}^{(k)}) \right)$$

| Method | NMSE @ K=16 | Error energy [%] | Params |
|--------|-------------|------------------|--------|
| ISTA   | -5.3 dB     | 29.5%            | 0      |
| FISTA  | -11.0 dB    | 7.9%             | 0      |

At K=16 iterations both methods are far from converged.

ISTA needs 200+ iterations for -20 dB.

Can we make 16 steps count?

*ISTA = Iterative Shrinkage-Thresholding Algorithm; FISTA = Fast ISTA (Beck & Teboulle, 2009)*

# Deep Unrolling: From Iterations to Layers

Key idea (Gregor & LeCun, 2010): unfold K ISTA iterations into a K-layer network and learn the update parameters from data.

## Classical ISTA — parameters fixed

$$\mathbf{z}^{(k)} = \mathbf{x}^{(k)} + (1/L) A^T (b - A \mathbf{x}^{(k)})$$

$$\mathbf{x}^{(k+1)} = S_{\theta}(\mathbf{z}^{(k)})$$

Parameters  $1/L$  and  $\theta = \lambda/L$  are fixed constants.

Fixed — no learning

## Unrolled LISTA — parameters learned

$$\mathbf{z}^{(k)} = W_x^{(k)} \mathbf{x}^{(k)} + W_y^{(k)} b$$

$$\mathbf{x}^{(k+1)} = S_{\theta_k}(\mathbf{z}^{(k)})$$

$W_x(k), W_y(k), \theta_k$  are learned per layer via BPTT.

~6 M parameters — learned end-to-end

MBDL principle: preserve the known algorithmic structure (gradient step + soft-threshold);  
learn only what improves finite-depth reconstruction. Training minimizes  $\|\mathbf{x}(K) - \mathbf{x}^*\|^2$  via BPTT.

BPTT = Backpropagation Through Time — end-to-end backpropagation through all K unrolled layers, as when training a recurrent network

# The MBDL Ladder: LISTA → ALISTA → HyperLISTA

**LISTA** — learn all matrices per layer, ~6 M params:

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta_k} ( \mathbf{W}_x^{(k)} \mathbf{x}^{(k)} + \mathbf{W}_y^{(k)} \mathbf{b} )$$

**ALISTA** —  $\mathbf{W}$  analytic, learn step sizes and thresholds, 32 params:

$$\mathbf{W}^* = ( \mathbf{A} \mathbf{A}^T )^{-1} \mathbf{A} \quad (\text{column-normalized})$$

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta_k} ( \mathbf{x}^{(k)} + \gamma_k \mathbf{W}^{*T} ( \mathbf{b} - \mathbf{A} \mathbf{x}^{(k)} ) )$$

**HyperLISTA** — adaptive  $\theta/\beta/p$  from residuals, 3 params (grid search, zero backprop):

$$\theta^{(k)} = c_1 \mu \parallel \mathbf{A}^+ ( \mathbf{A} \mathbf{x}^{(k)} - \mathbf{b} ) \parallel_1, \quad \beta^{(k)} = c_2 \mu \parallel \mathbf{x}^{(k)} \parallel_0$$

$$p^{(k)} = c_3 \log ( \parallel \mathbf{A}^+ \mathbf{b} \parallel_1 / \parallel \mathbf{A}^+ ( \mathbf{A} \mathbf{x}^{(k)} - \mathbf{b} ) \parallel_1 )$$

| Model             | Params   | NMSE            | Error [%]       |
|-------------------|----------|-----------------|-----------------|
| ISTA              | 0        | -5.3 dB         | 29.5%           |
| FISTA             | 0        | -11.0 dB        | 7.9%            |
| LISTA             | 6 M      | -20.3 dB        | 0.93%           |
| ALISTA            | 32       | -29.8 dB        | 0.10%           |
| <b>HyperLISTA</b> | <b>3</b> | <b>-61.4 dB</b> | <b>0.00007%</b> |

NMSE on synthetic sparse recovery ( $m=250$ ,  $n=500$ ,  $s=50$ , noiseless,  $K=16$  layers)

**More structure → fewer params → better performance**

This is the MBDL principle in action.

# Our Design Space Study: What Should We Learn?

Research question: LISTA learns 6 M parameters; HyperLISTA uses 3 scalars. What is the minimal useful set of learnable components?

The ISTA gradient-correction update has two free scalar sequences:

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta_k} \left( \mathbf{x}^{(k)} + \gamma_k A^T (b - A \mathbf{x}^{(k)}) \right)$$

We designed three models, each learning a different subset:

## ThresholdISTA

16 params

Learns  $\theta_1, \dots, \theta_K$  only

Step fixed at  $\gamma = 1/L$

Init:  $\theta_0 = \lambda/L$

## StepISTA

16 params

Learns  $\gamma_1, \dots, \gamma_K$  only

Threshold coupled:  $\theta_k = \lambda \gamma_k$

Init:  $\gamma_0 = 1/L$

## StepThresholdISTA

32 params

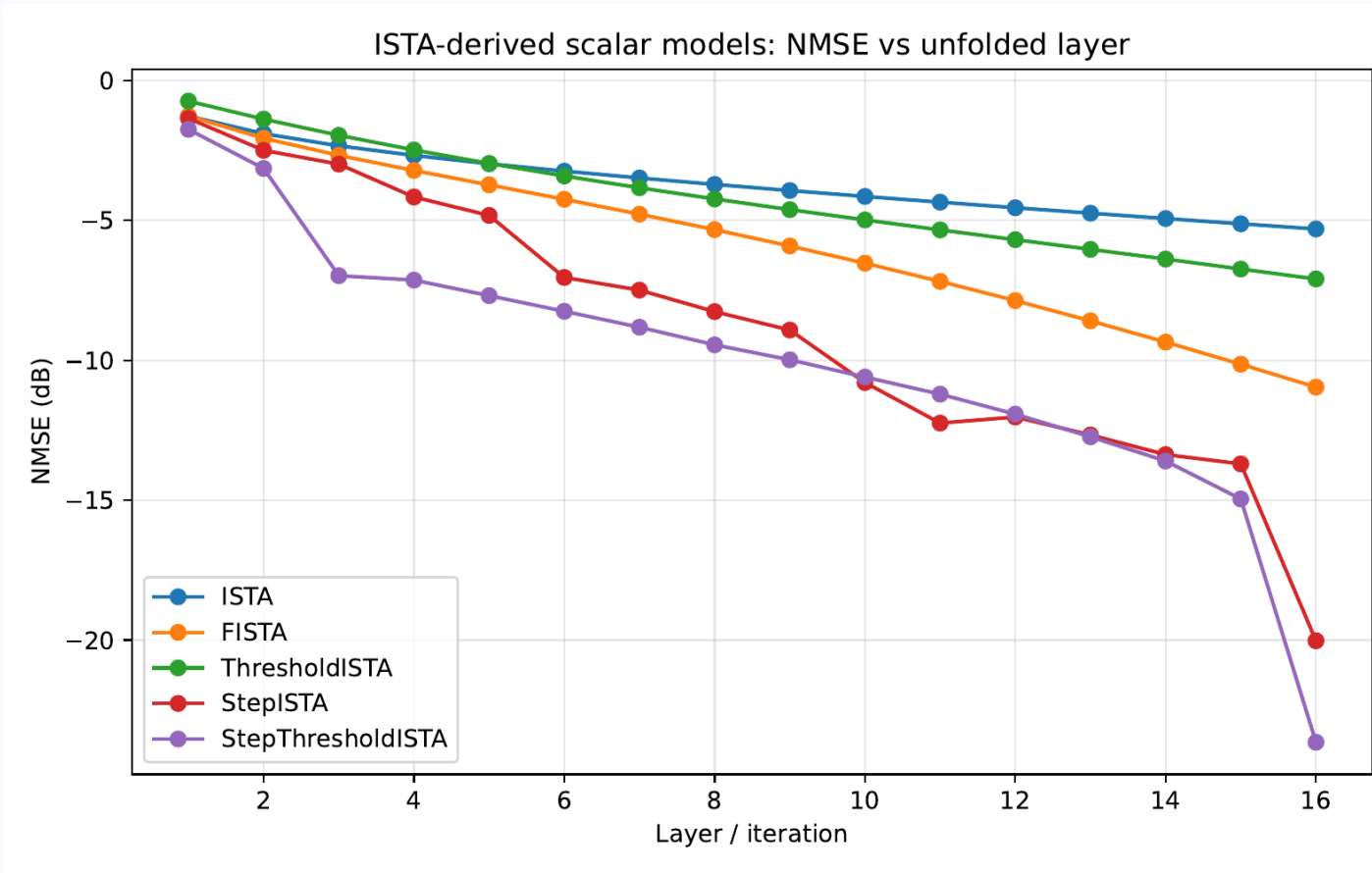
Learns both  $\gamma_k$  and  $\theta_k$

independently per layer

Init: ISTA values

*All parameters softplus-reparameterized (positivity). All initialized to reproduce ISTA exactly before training.*

# Our Results: Scalar Models



| Model             | NMSE     | Params |
|-------------------|----------|--------|
| ISTA              | -5.3 dB  | 0      |
| FISTA             | -11.0 dB | 0      |
| ThresholdISTA     | -7.1 dB  | 16     |
| StepISTA          | -20.0 dB | 16     |
| StepThresholdISTA | -23.6 dB | 32     |
| LISTA-L1 (ref.)   | -20.3 dB | 6 M    |

**Key finding**

Learning  $\gamma_k$  alone (16 scalars) already matches full LISTA (6 M parameters).

Decoupling  $\theta_k$  adds 3 dB at zero extra cost.

StepThresholdISTA beats LISTA-L1 with 187,000× fewer parameters.



# Training Methods for Unfolded Optimizers

Architecture fixed (LISTA). Only the training strategy changes.

**L1 — End-to-end**

$$L = \left\| \mathbf{x}^{(K)} - \mathbf{x}^* \right\|^2$$

Supervise final output only.  
Standard BPTT.

**L2 — Deep Supervision**

$$L = \sum_k w_k \left\| \mathbf{x}^{(k)} - \mathbf{x}^* \right\|^2$$

Supervise every  
intermediate layer.

**L3 — Greedy Layerwise**

$$\min_{\theta_k} \left\| \mathbf{x}^{(k)} - \mathbf{x}^* \right\|^2$$

Train layer k while  
layers 1...k-1 frozen.

## Tied vs. Untied weights:

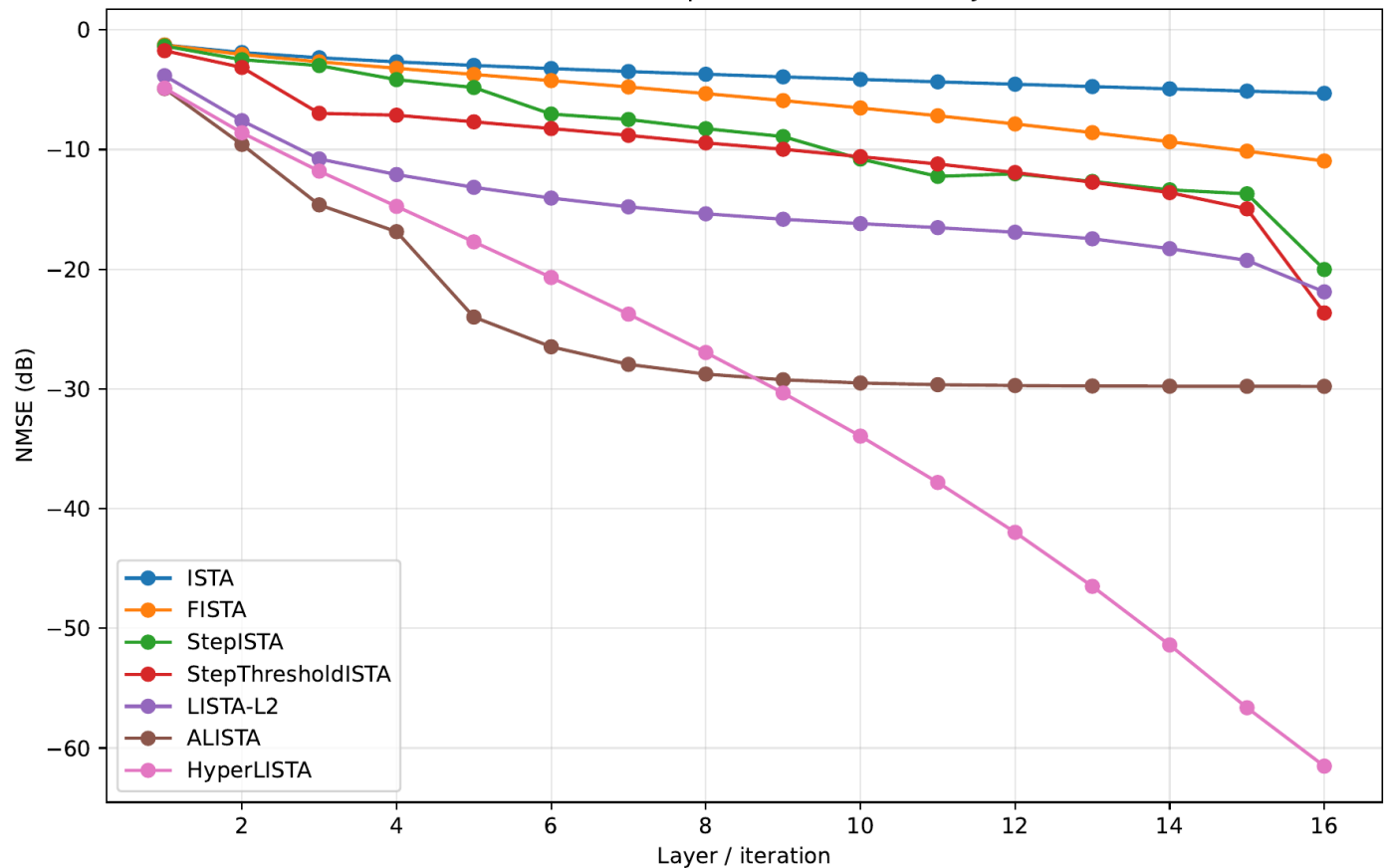
|         | Untied LISTA          | Tied LISTA                           |
|---------|-----------------------|--------------------------------------|
| Weights | Independent per layer | Single shared ( $W_x, W_y, \theta$ ) |
| Params  | ~6 M                  | 375 K                                |
| Style   | Fully flexible        | RNN-like                             |
| Effect  | Hard to optimize      | Smoother loss landscape              |

Weight tying = implicit regularization.

LISTA-Tied-L1: -19.1 dB @ 375 K  
LISTA-L1: -20.3 dB @ 6 M

# Synthetic Sparse Recovery Results

Main unfolded optimizers: NMSE vs layer

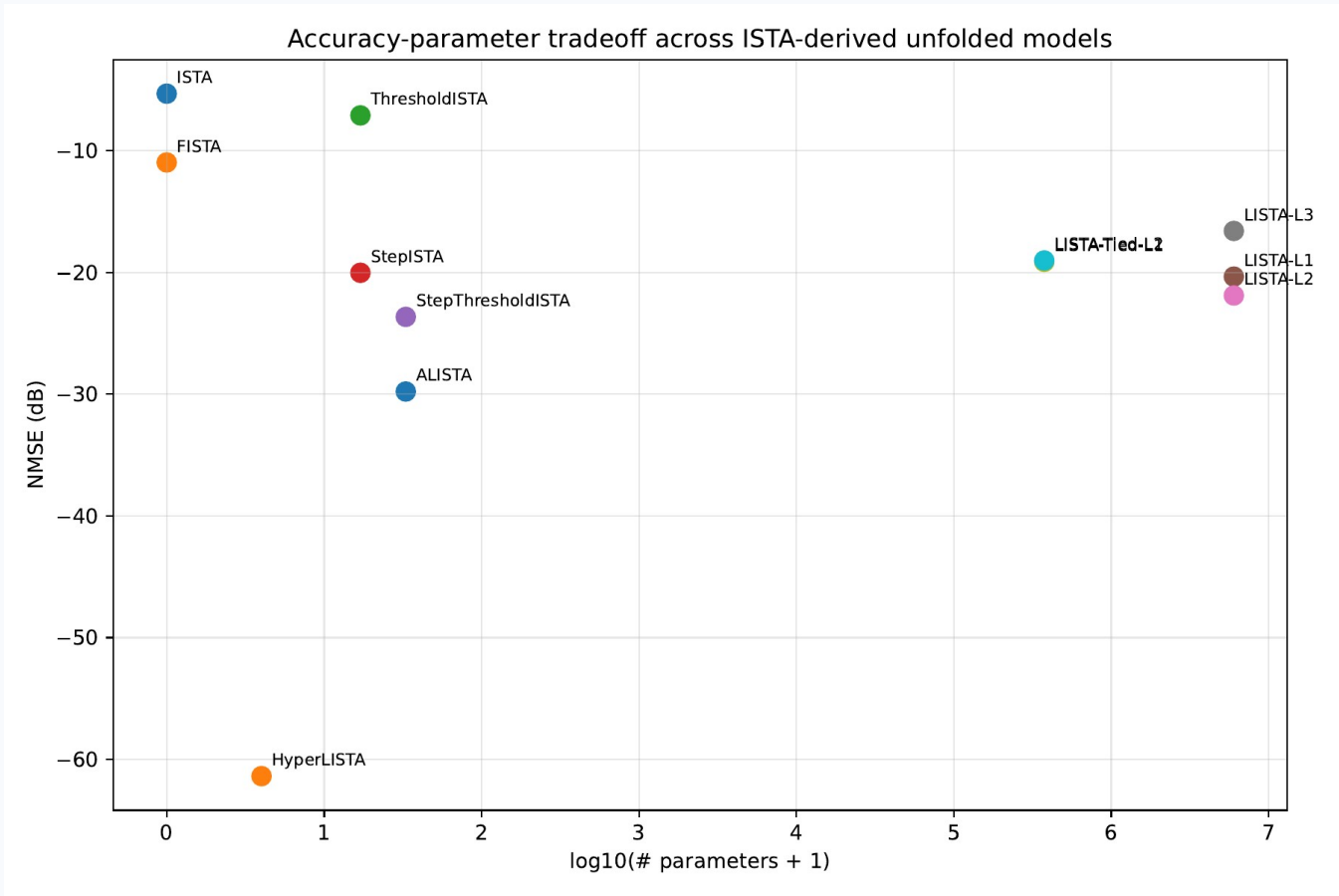


| Method        | NMSE     | Params |
|---------------|----------|--------|
| ISTA          | -5.3 dB  | 0      |
| FISTA         | -11.0 dB | 0      |
| LISTA-L1      | -20.3 dB | 6 M    |
| LISTA-L2      | -21.9 dB | 6 M    |
| LISTA-L3      | -16.6 dB | 6 M    |
| LISTA-Tied-L1 | -19.1 dB | 375 K  |
| ALISTA        | -29.8 dB | 32     |
| HyperLISTA    | -61.4 dB | 3      |

L3 plateau: NMSE flattens ~layer 8.  
Greedy training gives no incentive to improve beyond what prior layers solved.

LISTA-Tied > LISTA-Untied despite 16× fewer params.

# Design-Space Map



| Model             | Params | NMSE     | What drives it        |
|-------------------|--------|----------|-----------------------|
| ThresholdISTA     | 16     | -7.1 dB  | learned $\theta_k$    |
| StepISTA          | 16     | -20.0 dB | learned $\gamma_k$    |
| StepThresholdISTA | 32     | -23.6 dB | $\gamma_k + \theta_k$ |
| LISTA-L1          | 6 M    | -20.3 dB | full matrices         |
| ALISTA            | 32     | -29.8 dB | analytic W            |
| HyperLISTA        | 3      | -61.4 dB | adaptive residuals    |

**StepThresholdISTA (32 scalars)**  
**beats LISTA-L1 (6 million matrices).**  
The step-size schedule  $\gamma_k$  is the bottleneck.

# Real Image CS: FashionMNIST Pixel Domain

Setup: flatten 28×28 image to  $x \in [0,1]^{784}$ , measure  $b = Ax$ , ratio  $m/784 \in \{0.25, 0.5\}$ . FashionMNIST: ~51.5% of pixels exactly zero.

| Method            | NMSE            | PSNR        | SSIM        | Params      |
|-------------------|-----------------|-------------|-------------|-------------|
| ISTA              | −1.2 dB         | 8.9         | 0.13        | 0           |
| FISTA             | −1.2 dB         | 8.9         | 0.14        | 0           |
| StepThresholdISTA | −1.2 dB         | 8.9         | 0.13        | 32          |
| ALISTA            | −0.9 dB         | 8.4         | 0.12        | 32          |
| HyperLISTA        | −0.3 dB         | 7.7         | 0.09        | 3           |
| <b>LISTA-L2</b>   | <b>−13.7 dB</b> | <b>22.3</b> | <b>0.82</b> | <b>12 M</b> |

*ratio = 0.25*

## Signal model mismatch

All structured methods assume i.i.d.

Gaussian sparse signals.

FashionMNIST pixels are bounded  $[0,1]$ , spatially correlated, and only soft-sparse.

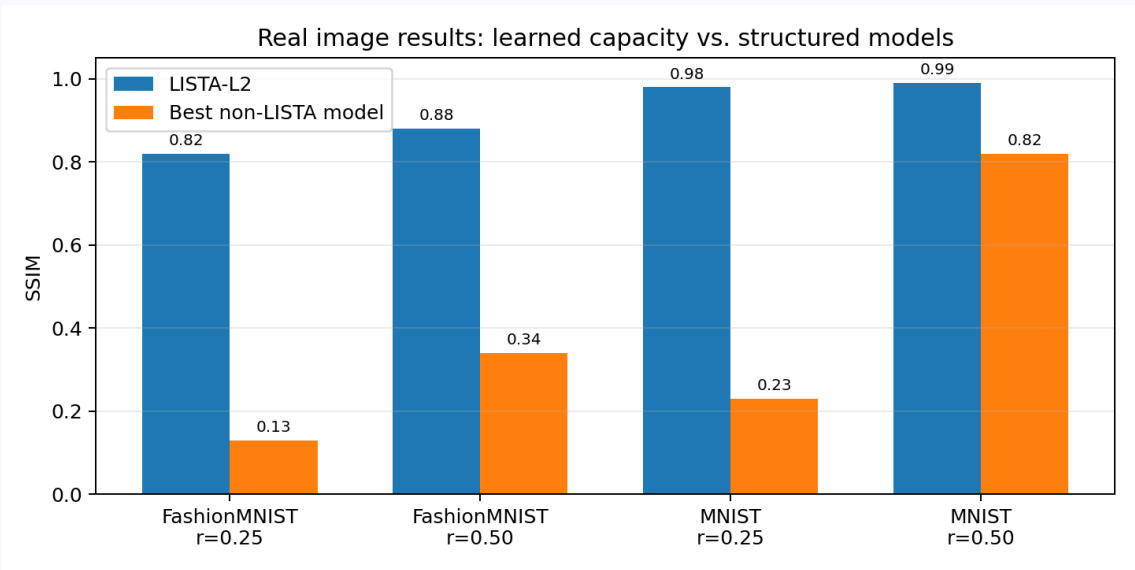
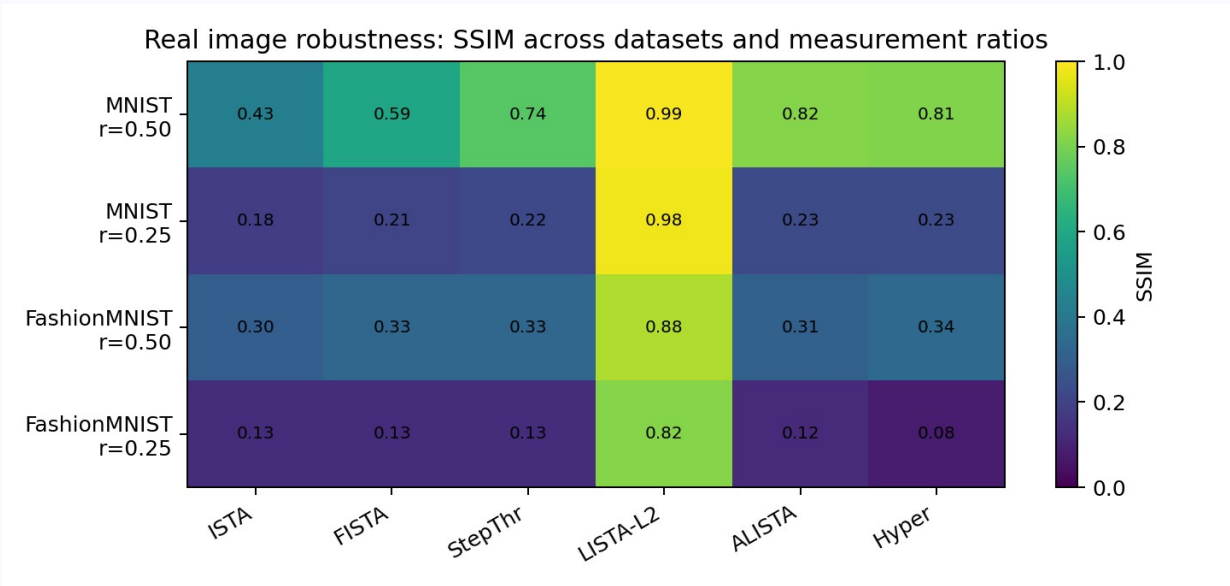
## Why LISTA wins

LISTA-L2 is assumption-free —  
it learns the actual image distribution  
from 51 K training images.

Wrong prior is worse than no prior.

# Robustness Across Datasets and Measurement Ratios

SSIM across {MNIST, FashionMNIST} × ratio {0.25, 0.5}: harder data and fewer measurements widen the gap.



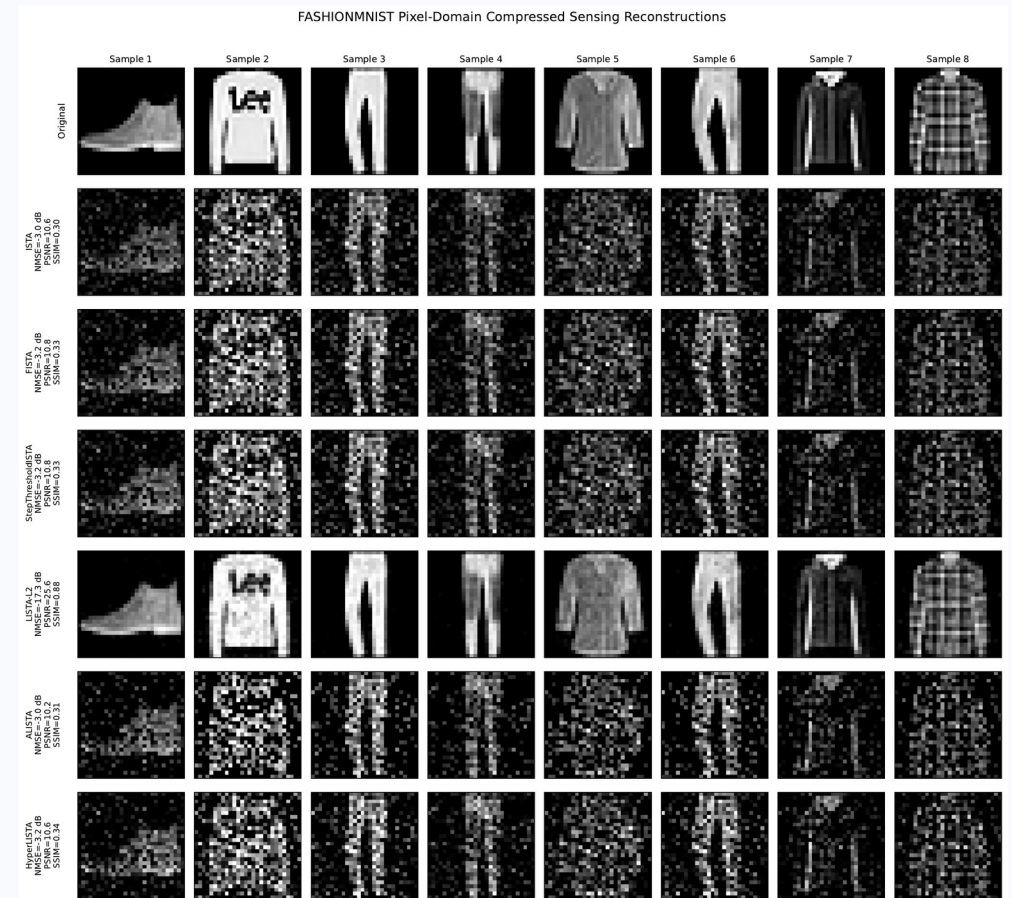
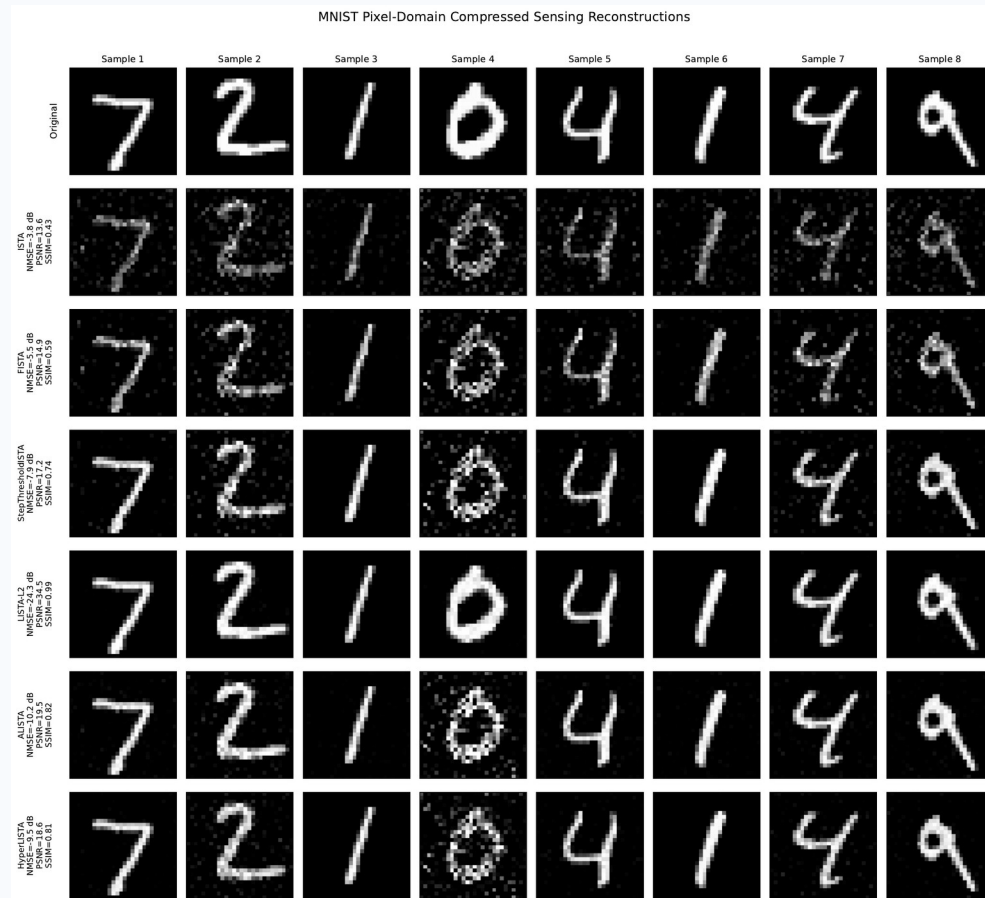
## Trend

MNIST @ 0.5 is easy for everyone; FashionMNIST @ 0.25 breaks every structured model. LISTA-L2 stays  $\geq 0.82$  SSIM everywhere.

## Interpretation

The sparse-Gaussian prior degrades gracefully on MNIST (~78% zeros) but fails on FashionMNIST (~51% zeros, textured).

# Visual Stress Tests (ratio = 0.25)



LISTA-L2 recovers clean structure on both datasets. ALISTA / HyperLISTA zero out valid signal on FashionMNIST — the Gaussian-sparse threshold is miscalibrated for textured images. Wrong prior is worse than no prior.

# Conclusions

## 1 Structure wins when assumptions match.

HyperLISTA's 3 scalars beat 6 M learned parameters by 41 dB on synthetic sparse data.

## 2 Step sizes matter most.

StepISTA (16 scalars) matches LISTA. Decoupling  $\theta_k$  adds 3 dB. Matrix learning is largely redundant.

## 3 Weight tying is implicit regularization.

LISTA-Tied matches LISTA-Untied (-19.1 vs -20.3 dB) with 16× fewer parameters.

## 4 Wrong prior actively hurts.

ALISTA / HyperLISTA score below ISTA on real images. LISTA wins by making no structural assumptions.

### MBDL hierarchy on synthetic data:

ISTA → LISTA → ALISTA → HyperLISTA

0 → 6M → 32 → 3 params

-5 → -20 → -30 → -61 dB

### Future work

Apply models in DCT / wavelet domain — images are sparser there, closer to the Gaussian-sparse model.

Design image-aware thresholds for bounded, non-negative pixel structure.

# Code & References

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**Code — public GitHub repository (notebooks 01-04 + src modules):**

[github.com/ThEpiCake/hyperlista-mbdl](https://github.com/ThEpiCake/hyperlista-mbdl)

## **References:**

- [1] A. Beck and M. Teboulle, "A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems," SIAM J. Imaging Sciences, 2009.
- [2] K. Gregor and Y. LeCun, "Learning Fast Approximations of Sparse Coding," ICML 2010.
- [3] X. Chen, J. Liu, Z. Wang, W. Yin, "Theoretical Linear Convergence of Unfolded ISTA and Its Practical Weights and Thresholds," NeurIPS 2018.
- [4] J. Liu, X. Chen, Z. Wang, W. Yin, "ALISTA: Analytic Weights Are As Good As Learned Weights in LISTA," ICLR 2019.
- [5] X. Chen, J. Liu, Z. Wang, W. Yin, "Hyperparameter Tuning is All You Need for LISTA," NeurIPS 2021.
- [6] N. Shlezinger and Y. C. Eldar, "Model-Based Deep Learning," Foundations and Trends in Signal Processing, vol. 17, no. 4, 2023.
- [7] Course lecture notes 1-8, Model-Based Deep Learning (361.2.2320), Ben-Gurion University, 2026.

**Thank you very much for listening!**